

DelightOn Garments Ltd.

Business Intelligence & Business Analytics - Specification Report

School-uniform manufacturer, Dublin, Ireland · Prepared for the BIBA project specification phase

1. Background and Business Goals

DelightOn Garments Ltd. is a Dublin-based manufacturer of school uniforms, supplying primary schools, secondary schools, Gaelscoileanna and a small number of colleges across Ireland. The business operates a made-to-order model: a school signs up for the academic year, pupils are measured on-site, garments are cut and stitched in DelightOn's own factory, and finished uniforms are delivered directly to the school ahead of term. It is a thin-margin, seasonally concentrated business - almost all volume has to land in the six to eight weeks before the academic year starts, which leaves very little room to absorb delay or rework once the season begins.

Over the past two academic years the company has moved from a period of steady, manageable growth into one of real operational strain. Enquiries are up sharply, but a shrinking share of them are turning into paying orders. Deliveries that used to arrive close to on time are now landing weeks late for the majority of schools. Fabric is being cut with far more waste than the factory plans for, and a meaningful share of what is purchased is never used at all. None of this shows up as a single dramatic failure - it shows up as a slow erosion across sales, operations and cash, which is precisely the kind of pattern that a properly instrumented CRM and BI layer is designed to catch early.

The marketplace itself is small and relationship-driven: schools typically re-order from the same supplier year after year, procurement decisions are made by a handful of staff per school, and price competition is intense because uniform budgets are set at household level. That makes retention and reliability, not just new customer acquisition, the metrics that actually determine whether DelightOn recovers. The business goals that follow directly from this are to rebuild the sales pipeline into something measurable and manageable, to restore delivery reliability to a level schools can plan around, to bring cutting-floor wastage back toward its planned rate, and to close the gap between what is invoiced and what is actually collected. Sections 4 and 5 of this report quantify each of these problems directly from DelightOn's own operational data, and the gap analysis below summarises how each finding maps onto a concrete BI or CRM response.

Competitively, DelightOn sits alongside a small number of other Irish school-uniform manufacturers and a growing number of online-only suppliers who compete almost entirely on price and turnaround time. Because most schools re-tender or at least reconsider their supplier every few years, the switching cost for a school is low, which means DelightOn's current combination of falling conversion and worsening lateness is not merely an internal inefficiency — it is actively creating an opening for competitors at exactly the schools DelightOn should be retaining. That competitive exposure is part of why the gap analysis below treats retention and reliability, rather than headline sales volume, as the primary lens for judging the CRM and BI intervention.

The wastage and idle-stock findings in Section 4 also connect directly to UN Sustainable Development Goal 12, Responsible Consumption and Production: a factory cutting fabric at more than double its planned wastage rate, with close to a fifth of purchased material never used at all, is not only a margin problem but a material-efficiency one, and the same BI instrumentation proposed here for cost control doubles as the reporting DelightOn would need to demonstrate genuine progress against that goal.

Gap Analysis

Gap identified	Evidence from the data	Planned BI / CRM response
Sales funnel leaking	Conversion fell from 27.1% to 15.5% while leads grew 57%; Social Media converts 0 of 14 leads	CRM pipeline stages, follow-up SLAs, source-level ROI reporting
Delivery reliability collapsing	86% of AY2026 orders were late (avg +15.2 days) versus 47% in AY2025; two vendors cause most late fabric	Vendor OTIF scorecard, reorder-point alerts, punctuality dashboard
Factory efficiency	Cutting wastage averages 11.2% against a 5% plan; 17.5% of purchased fabric sits idle in lots	Wastage Pareto by reason, lot-utilisation reporting
Cash & quality leakage	€68,992 of €514,905 invoiced remains uncollected; 4.8% of delivered pieces are returned	Receivables-ageing dashboard, automated reminders, defect tracking

Table 1 - Operational gaps identified in the data and the corresponding BI/CRM response.

2. System Architecture and Data Capture Flow

Before any dashboard or CRM object can be designed, it matters where DelightOn's data actually originates. The diagram below traces that lineage across the four physical points where information first enters the business: the sales office, the measurement visit at each school, the factory floor, and the delivery dock. Every table in the underlying database is generated by one of these four events, and each stage carries forward the keys of the stage before it, giving an unbroken thread from a school's first enquiry through to the individual garment that gets delivered to a named pupil.

A school enquiry is logged by a sales representative and classified by source and status; on conversion it becomes a customer account and an order. Staff then visit the school to take measurements pupil by pupil, which in turn drives fabric procurement, cutting, and a six-phase production log through to finished stock. Dispatch records one line per physical garment, capturing delivery and any returns, while invoicing and payments are tracked against the order throughout. All twenty-three operational tables ultimately feed the same analytics layer, which is what makes a single, consistent BI and CRM view possible rather than four disconnected departmental spreadsheets.

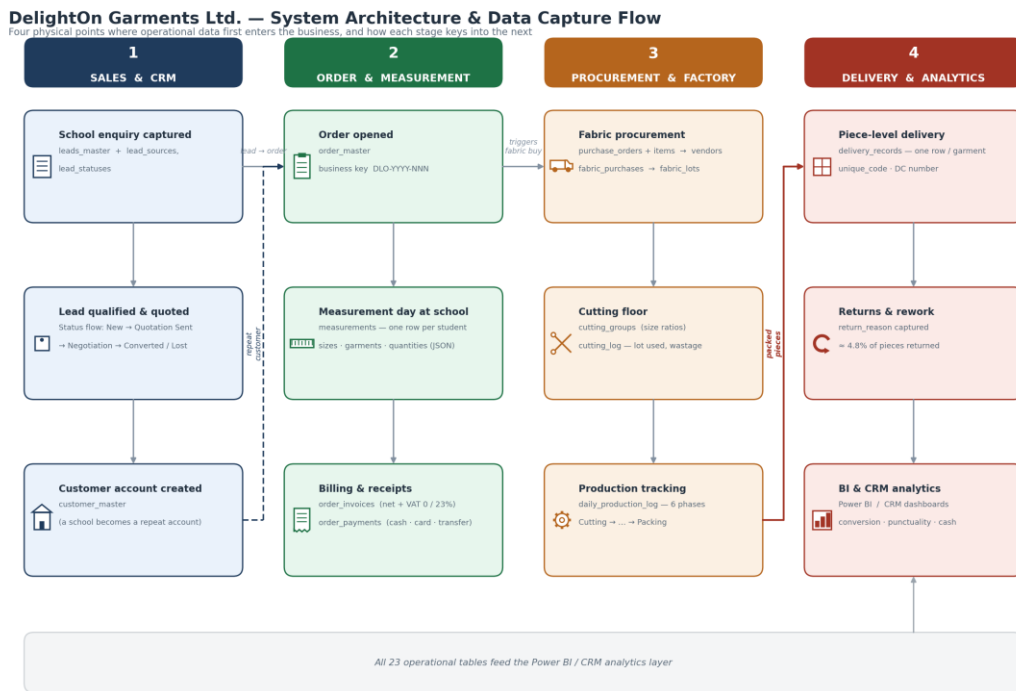


Figure 1 - System architecture and data capture flow, from first enquiry to delivered garment.

Customers are not passive in this flow: the school's own contact is the source of the measurement data, the approver of the quotation, and the recipient of delivery and billing communications, so the CRM implementation treats each school as an account with its own contact history, order history and outstanding balance visible in one place, rather than as a row in a spreadsheet reconstructed each time someone needs an answer. On the analytics side, the capture points above map directly onto the reporting layer DelightOn needs: a sales-funnel view driven by the CRM stage data, a punctuality and vendor-performance view driven by order and purchase-order dates, a factory-efficiency view driven by the cutting and production logs, and a cash and quality view driven by invoices, payments and returns - the four themes explored quantitatively in Section 4.

3. Database Design

The physical schema behind this flow runs to twenty-three operational tables, thirty-five primary relationships and sixty-six foreign keys once audit columns are included, which is more detail than is useful to read at a glance. The conceptual model below groups those tables into the five domains that actually matter for the business - CRM & Sales, Orders & Billing, Procurement & Fabric, Production, and Delivery & Stock - and shows only the relationships that cross between them, since those are the joins a dashboard or report will actually need to make.

Reading it left to right follows the business itself: a lead becomes an order, the order triggers a fabric purchase, fabric is cut and made up on the factory floor, and the finished pieces are shipped. Two relationships cut across that straight line - production is tied back to its originating order regardless of how long it takes to complete, and a delivery record ultimately fulfils the measurement taken for a specific pupil. The garment catalogue sits below the main flow as shared reference data, used by procurement, production and delivery alike. Full table-level attributes and all sixty-six relationships are documented separately in the data dictionary and schema.sql.

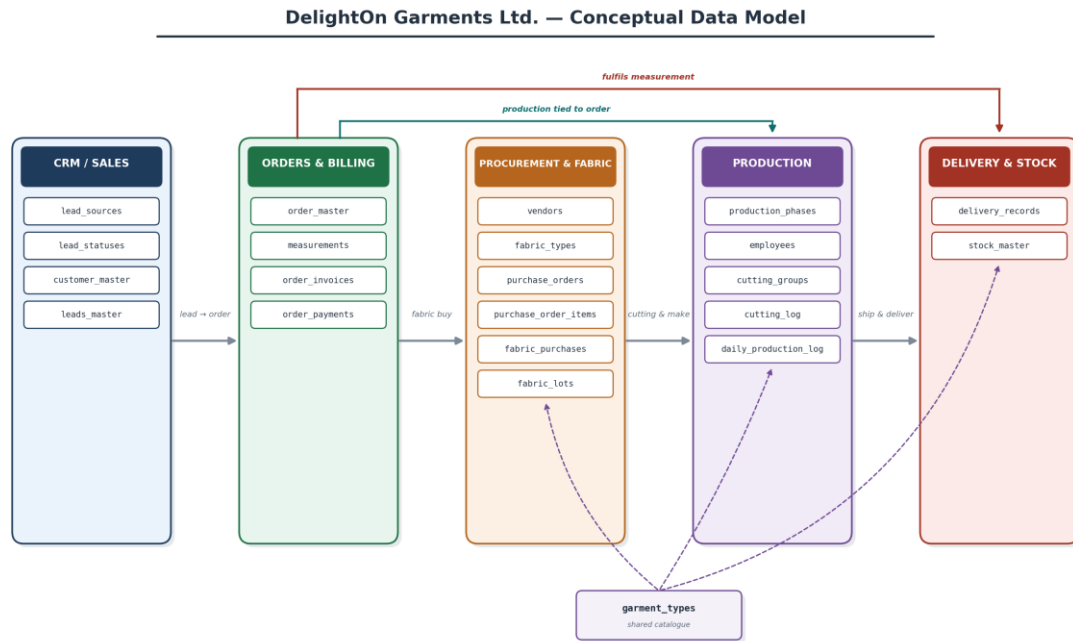


Figure 2 - Conceptual data model: the five operational domains and the primary flow of data between them.

A small number of design conventions run consistently through the physical schema and are worth stating explicitly, since they affect how any BI tool should treat the data. Every monetary column is stored in EUR with an explicit currency field alongside it rather than assumed from context; Irish VAT is calculated per line rather than looked up, at 0% for children's clothing and 23% for adult sizing and fabric; semi-structured data such as a student's measurements, garment list and quantities is stored as JSON rather than split across dozens of sparsely-populated columns, which keeps the schema stable as garment types change; and every table carries created/updated audit columns tied back to the user who made the change, which are omitted from Figure 2 for readability but are fully documented in the data dictionary. Order records use a human-readable business key (for example DLO-2026-014) alongside their numeric identifier, since two production tables - cutting groups and delivery records - reference the order by that business key rather than its surrogate, a deliberate exception that is called out wherever it occurs.

4. Exploratory Data Analysis

The statistics in this section come directly from DelightOn's own mock operational dataset - twenty-three tables and roughly thirteen thousand six hundred rows generated to mirror a real academic year of trading -and are reproducible from the schema referenced above. Read together, they are the evidence base for the gap analysis in Section 1: a sales funnel that is filling up but draining faster, a delivery promise that is being broken more often than kept, a factory that is burning more fabric than it plans to, and a cash position that is quietly slipping.

4.1 A funnel that is filling up but converting worse

Lead volume rose from 70 to 110 enquiries year on year, yet the share that convert into paid orders nearly halved, from 27.1% to 15.5%. That is not a demand problem - it is a follow-up and qualification problem, and the source breakdown makes clear where to focus first: education trade fairs and the website convert well above average, while Social Media has converted none of its fourteen leads and, more worryingly, the Repeat Customer channel - schools that have already bought from DelightOn before - converts at only 14.8%, well below what should be the warmest segment in the pipeline.

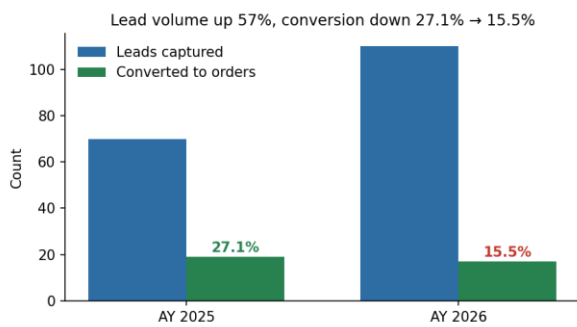


Figure 3 - Lead volume up 57%, conversion down 27.1% → 15.5%.

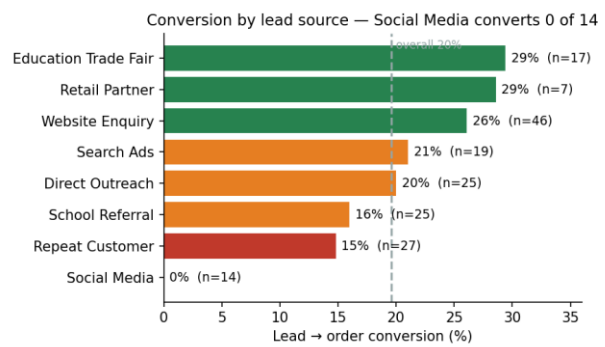


Figure 4 - Conversion by lead source; volume is not the same as quality.

4.2 A delivery promise that is being broken

Of the orders delivered in AY2025, 47% arrived late, by an average of 13.3 days; in AY2026 that has worsened to 86% late, by an average of 15.2 days. Fabric supply is a direct contributor: 39% of all purchase orders arrived late, and the problem is concentrated rather than diffuse - Quayside Fabric Traders and Boyne Valley Textiles account for the majority of it, at 67% and 50% of their respective orders running late. A vendor scorecard built on this data would let DelightOn act on two suppliers rather than guessing across eight.

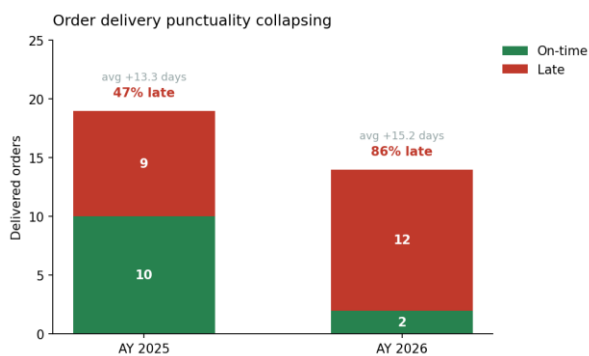


Figure 5 - Order punctuality has collapsed year on year.

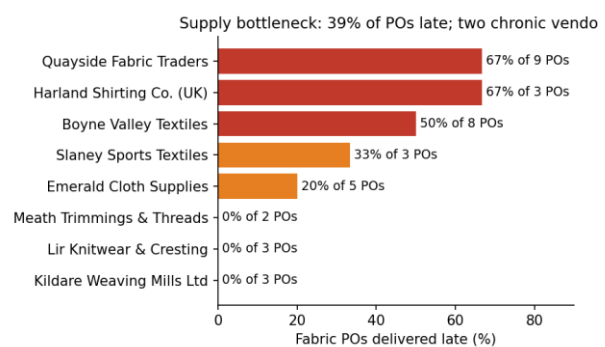


Figure 6 - Two vendors drive most of the late fabric supply.

4.3 A factory that is burning more fabric than it plans to

Across 194 logged cutting lays, actual wastage averages 11.2% against a planned rate of 5% - 2.4 times the target - with individual lays running as high as 34%. That spread points to a process issue rather than a material one: operator re-cuts, shade variation between fabric lots and pattern misalignment on repeat prints are the most frequently logged reasons, all of which are addressable through better marker planning and tighter lot-shade control rather than by buying different fabric. The same fabric-lot data shows that 17.5% of everything purchased - worth roughly €36,600 of the €209,214 total spend, is still sitting unused.

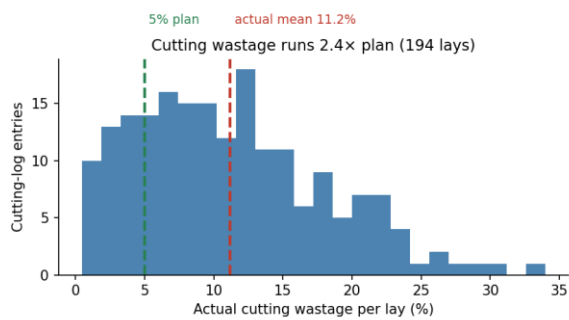


Figure 7 - Cutting wastage runs at 2.4× the planned rate.

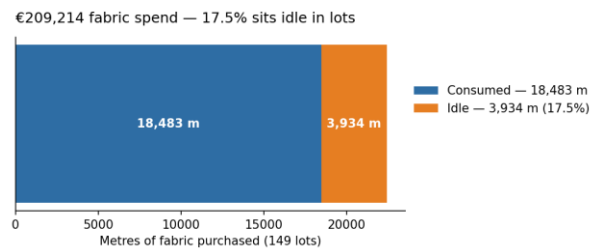


Figure 8 - 17.5% of purchased fabric remains idle in stock.

4.4 Cash and quality are both leaking quietly

Of €514,905 invoiced, €68,992 - around 13% - remains uncollected, and 4.8% of the 5,370 garments delivered have come back as returns, most commonly for size remakes, hem faults and shade mismatch. Neither figure is dramatic in isolation, but together they represent a meaningful, recurring drag on margin that a receivables-ageing dashboard and a defect-tracking view would make visible on an ongoing basis rather than only at year end.

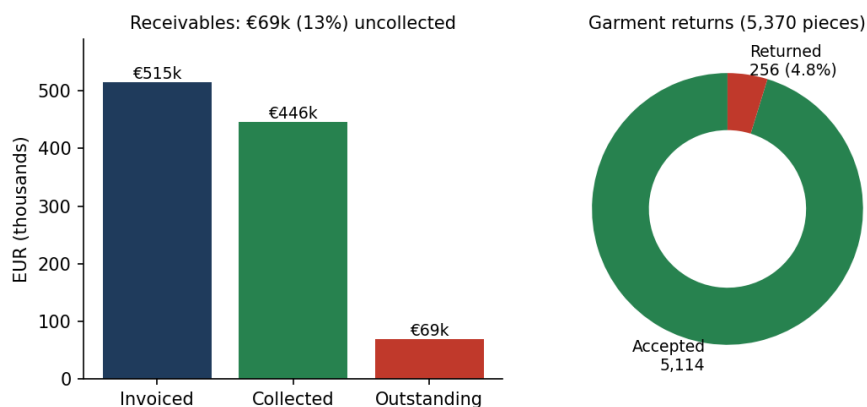


Figure 9 - Outstanding receivables and piece-level returns.

5. Data Pre-Processing

The dataset underpinning this analysis was synthesised, validated and load-tested before any of the figures above were produced, so that the statistics reported here reflect a structurally sound dataset rather than an artefact of messy inputs. Reference entities - users, employees, vendors, fabric and garment types, lead sources and statuses - were seeded first so every transactional row could key into a controlled vocabulary. All personal data was generated synthetically using an Ireland-locale faker library, so no real pupil, parent or staff data exists anywhere in the set. Every monetary figure was normalised to EUR and Irish VAT was calculated rather than randomised - 0% on children's clothing, 23% on adult sizing and on fabric purchases - with the identities grand total equals net plus VAT, and grand total equals advance plus payments plus balance, holding across all thirty-six invoices to within rounding error. Referential integrity was checked programmatically across all sixty-six foreign keys, including the two that reference the order's business key rather than its surrogate key, and the full schema was then executed against a live database with every CSV loaded under primary-key, foreign-key, unique and not-null constraints - all rows loaded without a single rejection, which is the strongest available check that the data and the structure genuinely agree.

6. Summary and Next Steps

Taken together, the architecture, database design and exploratory analysis in this report point to the same conclusion from three different directions: DelightOn's operational data is already detailed and trustworthy enough to build on, but nobody in the business can currently see it in one place. The conversion, punctuality, wastage and receivables figures in Section 4 are not new problems - they are problems the business has effectively been carrying for two academic years without a dashboard to surface them early. A CRM built around the school account, paired with the four analytical views set out in Section 2, would put each of these figures in front of the person who can act on it — the sales team, the buyer, the factory floor, and accounts - on an ongoing basis rather than only when a report happens to be pulled together.

The implementation phase that follows this specification will build the CRM objects and the Power BI dashboards against exactly the domains and relationships set out in Sections 2 and 3, and will report back against the same conversion, punctuality, wastage and receivables baselines established here, so that any improvement - or any part of the gap that proves harder to close than expected - can be measured against real, pre-implementation figures rather than an assumption.